Word Frequency in Classic Novels

In this workspace, you'll scrape the novel Moby Dick from the website Project Gutenberg (which contains a large corpus of books) using the Python `requests` package. You'll extract words from this web data using `BeautifulSoup` before analyzing the distribution of words using the Natural Language ToolKit (`nltk`) and `Counter`.

The Data Science pipeline you'll build in this workspace can be used to visualize the word frequency distributions of any novel you can find on Project Gutenberg.

```python
# Import and download packages
import requests
from bs4 import BeautifulSoup
import nltk
from collections import Counter
nltk.download('stopwords')

# Start coding here...
```

```
[nltk_data] Downloading package stopwords to /home/repl/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

```
True
```

```python
# Get the webpage
r =
requests.get('https://s3.amazonaws.com/assets.datacamp.com/production/project_147/da
tasets/2701-h.htm')

# Set encoding type
r.encoding = 'utf-8'

# Extract HTML text
html = r.text

# Preview first 2000 chars
print(html[0:2000])
```

```
<?xml version="1.0" encoding="utf-8"?>

<!DOCTYPE html
   PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
   "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd" >

<html xmlns="http://www.w3.org/1999/xhtml" lang="en">
  <head>
    <title>
      Moby Dick; Or the Whale, by Herman Melville
    </title>
    <style type="text/css" xml:space="preserve">

    body { background:#faebd0; color:black; margin-left:15%; margin-right:15%;
text-align:justify }
    P { text-indent: 1em; margin-top: .25em; margin-bottom: .25em; }
    H1,H2,H3,H4,H5,H6 { text-align: center; margin-left: 15%; margin-right: 15%;
}
    hr  { width: 50%; text-align: center;}
    .foot { margin-left: 20%; margin-right: 20%; text-align: justify; text-
indent: -3em; font-size: 90%; }
    blockquote {font-size: 100%; margin-left: 0%; margin-right: 0%;}
    .mynote    {background-color: #DDE; color: #000; padding: .5em; margin-left:
10%; margin-right: 10%; font-family: sans-serif; font-size: 95%;}
    .toc       { margin-left: 10%; margin-bottom: .75em;}
    .toc2      { margin-left: 20%;}
    div.fig    { display:block; margin:0 auto; text-align:center; }
    div.middle { margin-left: 20%; margin-right: 20%; text-align: justify; }
    .figleft   {float: left; margin-left: 0%; margin-right: 1%;}
```

```python
# Create soup object
html_soup = BeautifulSoup(html, "html.parser")

# Get plain text
moby_text = html_soup.get_text()
```

```python
# Create regex tokenizer
tokenizer = nltk.tokenize.RegexpTokenizer('\w+')

# Split into word tokens
tokens = tokenizer.tokenize(moby_text)
```

```python
# Convert all to lowercase
words = [token.lower() for token in tokens]

# Preview first eight
words[:8]
```

```
['moby', 'dick', 'or', 'the', 'whale', 'by', 'herman', 'melville']
```

```python
# Load English stop words
stop_words = nltk.corpus.stopwords.words('english')

# Preview first eight
stop_words[:8]
```

```
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all']
```

```python
# Filter out stop words
words_no_stop = [word for word in words if word not in stop_words]

# Preview first five
words_no_stop[:5]
```

```
['moby', 'dick', 'whale', 'herman', 'melville']
```

```python
# Count word frequencies
count = Counter(words_no_stop)

# Get top ten words
top_ten = count.most_common(10)

# Print the results
print(top_ten)
```

```
[('whale', 1246), ('one', 925), ('like', 647), ('upon', 568), ('man', 527), ('ship', 519), ('ahab', 517), ('ye', 473), ('sea', 455), ('old', 452)]
```